

Manual

Controlling of Shop Floor Data/Order Data BDE-CAB 8.1

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1 Übersicht Controlling Betriebsdaten / Auftragsdaten

Purpose

Business data controlling makes it possible to evaluate data entered in the system as concerns target/actual comparisons relating to quantities, times and dates.

Implementation considerations

You use the function package if:

- You would like to evaluate the on-time delivery performance of production based on the actual order / operation deadlines
- You would like to evaluate the times and durations recorded with reference to orders
- You would like to evaluate hours accrued in an area / a department over a period of time relating to various time accounts

Integration

The business data / order data monitoring function evaluates orders or operations that were either created directly in the system or that were transferred from other systems via interfaces.

Times and quantities entered in the BDE or MDE are accessed to display order progress.

Features

- Schedule controlling
 - Tabular evaluation of delayed order and early orders
- Order statistics
 - Order statistics with evaluations of finished and active orders, articles produced with target / actual comparison of quantities, performance and times, gross / net comparison to actual performance and remaining runtime, graphic and numeric evaluations of resource performance accounts.
- Overhead cost controlling
 - Evaluation of times recorded for overhead cost orders presented in a pivot table. Individual definition of the pivot table presentation by various criteria. Graphical presentation (bar chart). Predefined detail applications for displaying times for "activities per calendar week", "cost centers per calendar week", "cost centers per activity". Graphical display of the respective results.
- Maintenance controlling

- Evaluation of times entered for maintenance orders presented in a pivot table. Individual definition of the pivot table presentation by various criteria. Graphical presentation (bar chart). Predefined detail applications for displaying times for "articles per calendar week", "workplaces per calendar week", "articles per workplace". Graphical presentation of the respective results.
- Production controlling
 - Evaluation of times entered for production orders presented in a pivot table Individual definition of the pivot table presentation by various criteria. Graphical presentation (bar chart).
- Order profile
 - Order profile with display of times posted for operations (order duration, processing time, production and down time) incl. Gantt chart of calculated idle times.

2 Order Profile

1.1 Summary

Menu	Order Management --> Order Controlling --> Order Profile
Transaction code	orpf
Function authorization	orpf

Utilization

The order profile is indispensable for all production schedulers who want to see at a glance how orders have passed production. Due to its combination of table and Gantt chart, the production scheduler gets all required pieces of information at the push of a button.

Integration

The order profile is a report for production management

The order profile has been configured for the user to consider production orders that have already been realized. It responds to questions such as:

- Which orders have been produced (finished) within a certain period of time?
- Which output (quantities) have these orders produced?
- How much time was required?

Order controlling is made on order level and considers quantities and times with respect to the order (headers). Only finished orders are included as certain values (key figures) refer to the entire order. Consequently, the chronological selection refers to the order end.



Please note: This report is not ideal for “long-running orders” (serial orders) because of the selection option. However, the values and key figures displayed here are normally less interesting for serial orders.

Prerequisite

Only orders the order type of which is assigned to the category

- FA (production order),
- PJ (project order) or

- PM (plant maintenance order)

are displayed.

Selection Criteria

The application provides the following selection criteria:

Please take into consideration that in general - irrespective of the restrictions made using the selection criteria listed in the following – only orders that have already been started and the order type of which is assigned to the category

- FA (production order),
- PJ (project order) or
- PM (plant maintenance order)

are taken into account.

The application provides the following selection criteria:

Order

The "order" option restricts the displayed orders. Wildcards may be used.

Order type

If the order type is selected only those orders are taken into account that correspond to the selected order type. Multiple selections are possible.

Category

If the order category is selected only those orders are taken into account that correspond to the selected order type. This selection criterion refers to the category of the order type of the order. Only orders the order type of which corresponds to the specified category are displayed. Multiple selections are possible. Irrespective of this selection, only orders of the category "production order", "project order" and "maintenance order" are taken into account.

Article

When the article is selected, all orders are determined the article number of which at the order header matches the entered value. Wildcards are taken into account.

Article designation

If the article designation is selected all orders are determined the article designation of which at the order header corresponds to the entered value. Wildcards are taken into account.

Project number

If the project number is selected all orders are determined the project number of which at the order header corresponds to the entered value. Wildcards are considered.

Sales order

If the sales order number is selected all orders are determined the sales order number of which at the order header corresponds to the entered value. Wildcards are taken into consideration.

Customer designation

When the customer designation option is selected, all orders are determined the customer designation of which at the order header matches the entered value. Wildcards are considered.



The responsibility area is not checked in this application.

"Order profile" detail application (table)

All orders matching the entered selection criteria are displayed in the "order profile" detail application.

The displayed data is described in the sections that follow (the number of columns may vary subject to the column configuration):

Status

Order status (only orders are considered that have started or that are finished)

Order**Order, article, sales order**

Order header data

Target quantities**Target quantity, unit**

Target quantity of the order in base quantity unit.

Actual quantities**Yield, scrap, rework, open quantity, unit**

Order quantities in base quantity unit posted onto the order header. However, it is required that order backlog data that is manually created or transferred from the higher-tier system allows for the primary quantity unit to be converted into base quantity unit at operations.

Target times**Setup time**

Total of setup times (setup times+additional setup time+teardown time) of all active OPs according to the order backlog. The additional setup time results dynamically from using the setup change matrix in the HYDRA shop floor scheduling module.

Processing time

Total of processing times of all active OPs according to the order backlog.

Execution time

Total of setup time and processing time according to the order backlog.

Waiting time

Total of waiting times of all active OPs according to the order backlog.

Wait time

Total of wait times of all active OPs according to the order backlog.

Transport time

Total of transport times of all active OPs according to the order backlog.

Labor utilization

Total of target labor utilization times of all active OPs. The target labor utilization time needs to be transferred via the HYD-ERP interface.

Actual times**Retention period of orders**

The retention period of the order is the period between the time the order was first transferred from the PPS system ("order release" = order header creation date in HYDRA) and the time of the actual logoff (in terms of time) of the last active operation of the order.

Please note:

Whether the time of the order transfer is the time it was first transferred from the PPS system or the time it was re-transferred because it had been deleted in the meantime for technical reasons, cannot be determined or taken into consideration.

If an order is transferred more than once, and with each transfer the previous order is deleted, the creation date of the order header is the time of the last transfer.

Lead time

The order duration is the period between the time the first operation of the order is logged on and the time the last active operation is logged off

Processing time

The processing time of the order is the total of main utilization times (RPA 11) of all operations of this order.

Downtime

The downtime of the order is the total of downtimes (RPA 1 to 6, RPA 8 to 10) of all operations of the order.

Assignment time

The occupancy time is the total of setup time (RPA 7), processing time (RPA 11) and downtimes (RPA 1..6, RPA 8..10) of all operations of the order.

Wait time

The wait time results from times during which no operation was logged on. The duration is synchronized with the Gregorian calendar (shift breaks are not considered, durations posted onto RPA 12 "free break" are not considered; the posting times of the log records of the record type "U" and "E" are evaluated).



The wait time is not calculated at the time when data is requested, but by a separate process. For this purpose, however, this process needs to be integrated in the Scheduler. This procedure is described in the document entitled [Activating_OrderRelatedKeyfigures.pdf](#).

Labor utilization

The labor utilization of the order is the total of labor utilization times of all operations of the order.

Key figures

Rate of capacity utilization

The rate of capacity utilization is the ratio of processing time (RPA 11) to occupancy/assignment time (RPA 1..11) of all operations of the order

Setup rate

The setup rate is the ratio of setup time (RPA 7) to occupancy/assignment time (RPA 1..11) of all operations of the order.

"Order profile" detail application (graphic)

The article profile shows for each operation of the order selected in the table when the individual operation was logged on.

The operation number as well as the workplace, which the operation was logged on to, are displayed on the y-axis. When it comes to split OPs, the split number is shown in addition to the operation number. Data is displayed in ascending order by the operation number.

If there are simultaneous postings for an operation (e.g. as the operation has been logged on to several machines/workplaces at the same time) data is displayed in several lines for the operation.

The postings generated in HYDRA ("ADE log") are displayed and presented as bars (Gantt chart).

The times of resource performance accounts distributed in a posting are displayed within a bar according to the corresponding colors defined for the resource performance accounts. The sequence of RPA colors (ascending by RPA number) within a posting (of a bar), however, does not describe the sequence in which RPAs are posted, but remains the same for all postings (bars).

In case the duration (the time posted onto the RPA) within the log record is less than the time between logon and logoff, the remaining time is hatched within the order profile. This can be the case if

- the OP is part of a merged operation
- the OP was posted proportionately (this might, among others, also be the case for merged OPs (MOP) logged on to the terminal)

At the end of the graphic black bars represent wait times. These times result from the time between logging the operation off and logging the next operation on (the same operation or even different operations). This means, that during that time no operation of the order was logged on.

Tooltip

When going with the mouse pointer over a bar, a window opens for the resource performance account where the mouse pointer is currently located and the following information is displayed:

- - Current operation,
- Logon/logoff time of the operation
- - Yield (in primary quantity unit) as well as quantity unit
- Duration during the logon/logoff period
- Workplace, to which the operation was logged on
- Abbreviation and designation of the resource performance account as well as the duration posted onto this resource performance account



For HYDRA-MDE machines the operation is automatically interrupted when the shift ends and automatically logged on when the shift starts. Operations that are logged on when shifts are changing get two posting records and, as a result, 2 bars are displayed in the graphic.



The sequence of RPA colors within a posting (a bar) does not represent the order in which the RPA is posted, but remains the same for all posting records (bars).



If there are simultaneous postings for an operation (e.g. as the operation has been logged on to several machines/workplaces at the same time) data is displayed in several lines for this operation.



Representation of times (italic) on the y-axis is accidental.

"Duration" detail application (graphic)

The "duration" detail application shows the below-mentioned data:

- Lead time (dark blue)
- Assignment/occupancy time (yellow)
- Processing time (light green)
- Setup time (turquoise)
- Downtime (red)
- Wait time (black)

Further information on this data can be found in the section entitled "order profile detail application (table)".

The duration is displayed to the right of each bar in the format hh:mm:ss, instead of an x-axis labeling.

3 Order-Related Statistic

Summary

	→	
Menu	Order Management --> Order Controlling --> Order-Related Statistics	
Transaction code	orst	
Function authorization	orst	

Utilization

The order-related statistic is a detailed report about orders. It is also possible to evaluate quantities and times of the selected orders.

Integration

The statistic generated by the evaluation includes a list of orders and corresponding operations. The following is displayed for each operation:

- "Operations" detail application
Tabular overview of the operations pertaining to the selected order
- "Total quantities/total times/downtimes by resource performance accounts" detail application
Graphic overview of the quantities and times accrued on the order.
- "Performance and run time" detail application
Tabular and graphic presentation of run times and durations based on operations

Selection criteria

The application provides the following selection criteria:

Order

The order number can either be entered directly or using the search dialog.

Order type

This selection criterion refers to the order type at the order header. Only orders assigned to the selected order type(s) are displayed.

Category

This selection criterion refers to the category of the order type at the order header. Only orders assigned to the selected category(ies) are displayed.

Order status

This selection criterion refers to the order statuses at the order header. Only orders assigned to the selected order statuses are displayed.

Control

This selection criterion refers to the control indicator at the order header. Only orders assigned to the selected control indicators are displayed.

Article

This selection criterion refers to the article in the order header. All orders assigned to the selected article are displayed. Wildcards may be used.

Article designation

This selection criterion refers to the article designation defined in the order header. All orders assigned to the selected article designation are displayed. Wildcards may be used.

Sales order

This selection criterion refers to the sales order defined at the order header. All orders assigned to the selected sales order are displayed. Wildcards may be used.

Customer designation

This selection criterion refers to the customer designation defined in the order header. All orders including the selected customer designation are displayed. Wildcards may be used.

Project number

This selection criterion refers to the project number defined in the order header. All orders that are assigned to the selected project number are displayed. Wildcards may be used.

Planned order

This selection criterion refers to the planned order defined in the order header. Wildcards may be used.

Cost object

This selection criterion refers to the cost object defined in the order header. All orders of the selected cost object are displayed. Wildcards may be used.

Priority from ... to

This selection criterion refers to the priority defined in the order header. All orders assigned to the selected priority are displayed.

Basic start date from ... until

This selection criterion refers to the basic start date defined in the order header. Orders planned on or between the selected basic start dates are displayed only.

Basic end date from ... until

This selection criterion refers to the basic end date defined in the order header. Orders planned on or between the selected basic end dates are displayed only.

Order end from ... until

This selection criterion refers to the actual job end of the order header. Orders, which are dated on or between the selected order end date, are displayed only.

Order index from ... to

This selection criterion refers to the order index defined in the order header. All orders having the selected order index are displayed.

Order group

This selection criterion refers to the order group defined in the order header. All orders that are assigned to the selected order group are displayed.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. All orders of the selected MRP controller are displayed.

If several selection criteria are used overlapping results are displayed in the order overview.



The responsibility area is not checked in this application.

"Order overview" detail application

The "order overview" detail application shows all orders that have been selected in the selection panel. Every row shows one order. The numerous columns, which may be hidden or displayed, show all available pieces of information that are summarized in reasonable categories.

The list shows specific order data. This includes [order backlog data](#) as well as current [status information on the order](#).

"Operations" detail application

The "operations" detail application shows all operations pertaining to the above selected order at a glance. If several orders are selected in the order overview, all operations of these selected orders are displayed. In this case, we recommend showing the "order" column.

The list shows specific operation data. This includes [data about the pool of operations](#) as well as current [status information on the operation](#).

“Total quantities” detail application

Subject to the order selected in the order overview, this detail application shows the overall collected quantities of the order. If several orders are selected the total quantities of the selected orders are displayed in a graphic.

"Total times" detail application

Subject to the order selected in the order overview, the overall collected times of individual operations of the order are displayed in this detail application. If several orders are selected the totals of the selected orders are displayed in a graphic.

“Downtimes by resource performance accounts” detail application

Subject to the order selected in the order overview, this detail application shows the overall recorded downtimes of the order separated by resource performance accounts. If several orders are selected the totals of the selected orders are presented in a graphic.

“Performance and run time” detail application

The “performance and run time” detail application compares the actual net and gross activities with the target activities, depending on the operations selected in the “operations” detail application.

Further information on the calculation of individual key figures can be found [here](#).

“Target/actual comparison” detail application

When target/actual values are displayed and compared in a table, the target values are compared with the recorded actual values of the selected operations. The following values are displayed:

- Yield
- Scrap
- Setup time
- Cycle time
- Partitioning
- Production time per piece (time/piece)

Further information on the calculation of individual actual values can be found [here](#).

“Activities” detail application

The “activities” detail application compares target activities with the rendered actual activity (gross and net) of selected operations in a graphic.

Further information on the calculation of individual key figures can be found [here](#).

"Durations" detail application

The “durations” detail application compares the order duration, production duration and downtimes of the selected operations in a graphic.

“Downtimes by resource performance accounts” detail application

The “downtimes by resource performance accounts” detail application compares the accrued times of the selected operations, separated onto resource performance accounts in a graphic.

4 Overhead Cost Controlling

Summary

Menu	Order management → Order controlling → Overhead cost controlling
Transaction code	ohcon
Function authorization	ohcon

Usage

A portion of costs created in a company that should not be ignored are the result of so-called overhead costs. The objective of the evaluation "Overhead cost controlling" is to provide a means to make these overhead costs transparent and to identify the real "cost monsters", while in doing so finding ways to introduce countermeasures that will help lower overall costs.

Generally, it is the responsibility of the cost center managers to conduct audits and analyses of this kind and to derive the relevant measures based on the results. In production, for example, this is the responsibility of the foremen.

In addition to evaluating overhead costs relating to a specific cost center, it is also important to determine which activities incurred these costs. Overhead cost orders can be used to illustrate a breakdown of this kind.

Definition of overhead costs:

"Costs that cannot be attributed to any specific product or performance unit (cost object, cost center), such as lease or rent payments, executive salaries. [...] Overhead costs are such costs that cannot be attributed to any allocation base directly."

(Source: <http://www.wirtschaftslexikon24.net/d/gemeinkosten/gemeinkosten.htm>)

Integration

Database used to evaluate the order data logs .

Requirement

In order for an evaluation to be meaningful, what is required is that the employees record their overhead cost times correctly and allocate them to the cost object appropriately (correctly prepared overhead costs order).

Selection criteria

The application provides the following selection criteria:

Cost center

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the cost center entered.

Workplace

The postings for the record type "U"/ "E" are selected that are posted to workplace entered for which the user has been authorized via the responsibility area of the workplace.

Group

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the group entered.

Period

The time period entered restricts the selection by log records. Such log records are selected that have a start date within the defined period.

Responsibility area

The responsibility area that is entered restricts the requested data to the defined responsibility area for the particular machine/ workplace.

Company

By using the selection by company, only the data records for the relevant company are displayed or are included in the evaluation.

Order

The order number can be entered or selected. The order number defined here (order header number) restricts the evaluation to the selected order.

OP designation

Only the recorded data entered for an operation defined in the field OP designation with the selected designation is used for the evaluation (free text).



The responsibility area is not checked in this application.

Field descriptions**Responsibility area**

The responsibility area defined at the machine for which the data was entered.

Date

The dates shown are presented broken down by day.

Workplace

By integrating the field workplace, the data can be distributed or grouped by the workplace at which the overhead costs are recorded.

Year - quarter - month - calendar week - weekday - day

This field provides the ability to group or distribute the displayed data respectively.

Group

The data shown is distributed and displayed based on the machine group that is defined at each machine/ at each workplace.

Cost center

This field allows the data to be distributed based on the cost center defined at the machine/ workplace.

Company

This field allows the data to be distributed based on the company defined at the machine/ workplace.

Order

The data shown is grouped or displayed based on the order number (order header number).

Article

Filtering by article makes it possible to filter the article number of the separate operations.

MES order number

As opposed to the order number, the MES order number is formed by combining the order number and the operation number. Therefore, the data is displayed grouped by the separate operation numbers.

OP designation

For each operation there is an operation designation that is defined in the master data at the operation.

Total duration

The sum total of all durations for the selected data records.

General detail applications

The evaluation provided in the overhead costs controlling considers the operations in the overhead costs orders category.

Target times

These relate to target times/ periods defined in the order backlog. The target times are not related proportionately to the selection period, but instead are attributed absolutely to the entire operation.

Setup time

Setup time + dismantling time + dyn. setup time

Execution times

Setup time + processing time

Actual time (definitions)

Setup time

RPA 7

Processing time

RPA 11

Execution times

Setup time + processing time

Downtime

RPA 1..6, RPA 8..10

Total duration ("Occupancy/assignment time")

Setup time + processing time + downtime time

Activities per calendar week detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per calendar week. The total durations per operation designation are presented in the lines as totals.

Cost center per calendar week detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per calendar week. The total durations per cost center are presented in the lines as totals.

Cost center per activity detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per operation designation. The total durations per cost center are presented in the lines as totals.

5 Production Controlling

Summary

Menu	Order management → Order controlling → Production controlling
Transaction code	pdcon
Function authorization	pdcon

Usage

In this evaluation, you can evaluate data from the production orders using criteria that you can define in advance.

Integration

In this evaluation, you can evaluate posting data for orders based on various criteria that can be defined in the pivot table.

Initially, the duration of the order is displayed per machine and calendar week. This display can be changed by modifying the data field combination.

Selection criteria

The application provides the following selection criteria:

Article

Restricted to articles

Article designation

Restricted to article designation

Cost center

The postings for the record type "U"/ "E" are selected that are posted to workplace for which the user has been authorized via the workplace's responsibility area, and which are assigned to the cost center entered.

Period

The time period entered restricts the selection by log records. The log records are selected that have a start date within the defined period.

Responsibility area

The responsibility area that is entered restricts the requested data to the defined responsibility area for the particular machine/ workplace.

Workplace

The postings for the record type "U"/ "E" are selected that are posted to the workplace entered for which the user has been authorized via the responsibility area of the workplace.

Group

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the group entered.

Company

By using the selection by company, only the data records for the relevant company are displayed or are included in the evaluation.

Order

The order number can be entered or selected. The order number defined here (order header number) restricts the evaluation to the selected order.

OP designation

Only the recorded data entered for an operation defined in the field OP designation with the selected designation is used for the evaluation (free text).



The responsibility area is not checked in this application.

Field descriptions**Responsibility area**

The responsibility area defined at the machine for which the data was entered.

Date

The dates shown are presented broken down by day.

Workplace

By integrating the field Workplace, the data can be distributed or grouped by the workplace at which the production order costs are recorded.

Year - quarter - month - calendar week - weekday - day

This field provides the ability to group or distribute the displayed data respectively.

Group

The data shown is distributed and displayed based on the machine group that is defined at each machine/ at each workplace.

Cost center

This field allows the data to be distributed based on the cost center defined at the machine/workplace.

Company

This field allows the data to be distributed based on the company defined at the machine/workplace.

Order

The data shown is grouped or displayed based on the order number (order header number).

Article

Filtering by article makes it possible to filter the article number of the separate operations.

Article designation

Even if several articles have the same article number, these can be differentiated by article designation. This way the data records displayed can be grouped by article designation.

MES order number

As opposed to the order number, the MES order number is formed by combining the order number and the operation number. Therefore, the data is displayed grouped by the separate operation numbers.

OP designation

For each operation there is an operation designation that is defined in the master data at the operation.

OP

The field OP (Operation) lists the operation number only (without the order header number, e.g. 0010). This allows you to group data records which may be from different orders, however which have the same operation number.

Total duration

The sum total of all durations for the selected data records.

Tools

By running a selection by tool, only those posting records are used in the evaluation that were recorded for operations, in which the relevant tool was defined.

Pivot table detail application

You can evaluate order data based on additional criteria in the pivot view detail view.



The bar colors in the chart are set "arbitrarily" using a color chart defined internally.

6 Maintenance Controlling

Summary

Menu	Order management → Order controlling → Maintenance controlling
Transaction code	pmcon
Function authorization	pmcon

Usage

Maintenance controlling is a production management function. Maintenance especially is provided an overview of the maintenance orders that need to be processed.

The purpose of maintenance controlling is to show costs (based on activities/ times) that were incurred as a result of maintenance activities. By structuring the maintenance orders accordingly referencing the maintained equipment, the function provides the ability to identify maintenance-intensive materials.

Integration

Database used to evaluate the order data logs .

Selection criteria

The application provides the following selection criteria:

Article

Restricted to articles

Article designation

Restricted to article designation

Cost center

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the cost center entered.

Responsibility area

The responsibility area that is entered restricts the requested data to the defined responsibility area for the particular machine/ workplace.

Period

The time period entered restricts the selection by log records. Such log records are selected that have a start date within the defined period.

Workplace

The postings for the record type "U"/ "E" are selected that are posted to workplace entered for which the user has been authorized via the responsibility area of the workplace.

Group

The postings for the record type "U"/ "E" are selected that are posted to workplaces for which the user has been authorized via the workplace's responsibility area, and which are assigned to the group entered.

Company

By running the selection by company, only the data records are displayed or used in the evaluation that were created for the machines/ workplaces at those companies that match the companies you selected.

Order

The order number can be entered or selected. The order number defined here (order header number) restricts the evaluation to the selected order.

OP designation

Only the recorded data entered for an operation defined in the field OP designation with the selected designation is used for the evaluation (free text).



The responsibility area is not checked in this application.

Field descriptions**Responsibility area**

The responsibility area defined at the machine for which the data was entered.

Date

The dates shown are presented broken down by day.

Workplace

By integrating the field Workplace, the data can be distributed or grouped by the workplace at which the overhead costs are recorded.

Year - quarter - month - calendar week - weekday - day

This field provides the ability to group or distribute the displayed data respectively.

Group

The data shown is distributed and displayed based on the machine group that is defined at each machine/ at each workplace.

Cost center

This field allows the data to be distributed based on the cost center defined at the machine/ workplace.

Company

This field allows the data to be distributed based on the company defined at the machine/ workplace.

Order

The data shown is grouped or displayed based on the order number (order header number).

Article

Filtering by article makes it possible to filter the article number of the separate operations.

Article designation

Even if several articles have the same article number, these can be differentiated by article designation. This way the data records displayed can be grouped by article designation.

MES order number

As opposed to the order number, the MES order number is formed by combining the order number and the operation number. Therefore, the data is displayed grouped by the separate operation numbers.

Total duration

The sum total of all durations for the selected data records.

Tools

By running a selection by tool, only those posting records are used in the evaluation that were recorded for operations, in which the relevant tool was defined.

Article per calendar week detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per calendar week. The total durations per article number are presented in the lines as totals.

Workplace per calendar week detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per calendar week. The total durations per article number and workplace are presented in the lines as totals.

Article per work place detail applications

The data is presented in a pivot grid (top) and as a bar chart (bottom). The columns show the total duration added up per workplace. The total durations per article number are presented in the lines as totals.

7 Schedule Controlling

Summary

Menu	Order Management → Order Controlling → Schedule Controlling
Transaction code	scec
Function authorization	scec

Usage

This application is meant to be used as a monitoring list so that you can determine for each operation any downstream deviations in the schedule. It provides answers to the following questions, for example:

- Which operations with an exceeded planned start have not (yet) begun?
- Which operations are currently not being processed (status prepared or interrupted) and are expected to be delayed as a result?
- Which operations that are currently being processed (status running), are expected to be finished with a delay?
- Which operations were finished with a delay?

This function not only allows you to analyze delays in the schedule, but also which order start or which order end are running early.

This function is geared towards production controllers or foremen, for example. The benefit is that the function continuously monitors the quality of the planned times and assures production runs smoothly.

Integration

During the analysis, current dates are compared to planned dates. Depending on the type of planning, the function can check specific dates planned in detail, but also scheduled dates or basic dates. The planned dates here are either the result of detailed planning performed in HYDRA shop floor scheduling or from the planning performed in the ERP system.

Prerequisite

In order to make the best use of this application, you should be knowledgeable about how the planned dates are determined and set in your system so that based on this knowledge you can make better decisions about how to perform a selection.

Selection criteria

The application provides the following selection criteria:

Check against baseline plan

The selection list shows all baseline plans for which the user is authorized by the responsibility are authorization.

OPs started too early, OPs started too late, OPs finished too early, OPs finished too late

Depending on the selection options you chose, what is considered here are:

- Operations started too early
- Operations finished too early
- Operations started too late
- Operations finished too late

The values are calculated based on which option was selected for "Exceeding of planned dates", "Exceeding of basic dates" or "Exceeding of scheduled dates".

If none of these options is set, all operations are displayed that do not have any deviations.

Date ... to ...

Only those operations are checked that have a start date or a finish date (depending on the selection option) in this period.

Consider long-term data

If this option is set, operations that were already transferred from the online database into the long-term pool of data are also considered.

Exceeding of planned dates

The start and finish dates planned in detail during detailed scheduling (e.g. in HYDRA shop floor scheduling) should be checked to assure they can be maintained.

Exceeding of basic dates

The operation related basic dates *Latest start (LST)* and *Latest end (LET)* specified by the ERP system or those resulting from lead time scheduling in HYDRA are checked to assure they can be maintained.

Please note:

These basic dates at the operation that result from lead time scheduling must be transferred from the ERP system correctly. Alternately, there is also the option to run the basic dates through lead time scheduling in HYDRA in order to calculate the values.

Exceeding of scheduled dates

The dates Scheduled start time and Scheduled end time transferred from the ERP system or calculated in HYDRA in lead time scheduling should be checked to assure they can be maintained.

Planned on

Workplace or group

Workplace/ group/ cost center

Narrows down the display by workplace, group or cost center.

Please note:

The selection by workplace or cost center only makes sense for operations that have already been specifically scheduled at a workplace.

Order

This option allows you to limit your search to certain orders. You can also run a search using wildcards.

Article

Limiting the search to operations of a specific article. You can also run a search using wildcards.

Customer designation

Limiting the search to operations of orders for a specific customer. You can also run a search using wildcards.

OP designation

Limiting the search to certain operations.

Operation status

Limiting the search to operations of a specific operation status.

Field descriptions

After the data is requested, in addition to the order/ operation, OP designation and article the table also displays the basic dates, the dates of detailed planning and the actual dates.

Furthermore, schedule deviations (too early, too late, on-time delivery performance – see below) are displayed in the format days.hours:minutes:seconds.

If configured accordingly during customizing, customer-specific user fields are also shown for the order header or the operation .

Calculating data

The verification is performed against the planned dates/ scheduled dates/ basic dates (depending on the option selected) from the current pool of data.

The values are calculated based on

- which index tab "planned dates", "scheduled dates" or "basic dates" was selected.

The values "planned start" or "planned finish" described in the following table should therefore be considered placeholders for the various combinations (current planned dates/ planned dates, current planned dates/ scheduled dates, current planned dates/ basic dates).



The schedule deviations are calculated and displayed in the days.hours:minutes:seconds format, compared to the Gregorian calendar.

Beginning of deviations

Earliness

Condition 1: Control	Condition 2: Date	Calculation
L, U, E, A	OP actual start < planned start	Abs (OP actual start minus planned start)

Delay

Condition 1: Control	Condition 2: Date	Calculation
L, U, E, A, F	OP actual start > planned start	Abs (OP actual start minus planned start)
S, V	OP planned start > today	Abs (OP planned start minus today)

On-time delivery

On-time delivery = earliness + delay

End of deviations

Earliness

Condition 1: Control	Condition 2: Date	Calculation
E, A	OP actual end < planned end	Abs (OP actual end minus planned end)

Delay

Condition 1: Control	Condition 2: Date	Calculation
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Condition 1: Control	Condition 2: Date	Calculation
S, V, L, U, F	Today > planned end	Abs (today minus planned end)
E, A	OP actual end > planned end	Abs (OP actual end minus planned end)

On-time delivery

On-time delivery = earliness + delay

Detail application: Schedule controlling (graphic)

The detail application “schedule controlling (graphic)” shows the earliness, delay and on-time delivery for all operations selected in the table. If no operation is selected, it will be interpreted as “all operations selected”.

Display options

Display	Relating to	Group by	Consideration
☒ Delays	☐ Start	☒ Workplace	☒ Mean value/standard deviation
☐ Earliness	☒ End	☐ Group	☐ Total of durations
☐ On-time delivery		☐ Cost center	

Display:

Defines whether the graphic shows earliness, delay or on-time delivery performance.

Relating to:

Specifies whether the deviations to be considered are to refer to the beginning (planned start/actual start) or to the end (planned end/actual end) of the operation.

Group by:

Subject to the selection made in the selection panel, data are summarized and displayed by workplace, group or cost center.

Consideration:

Specifies whether

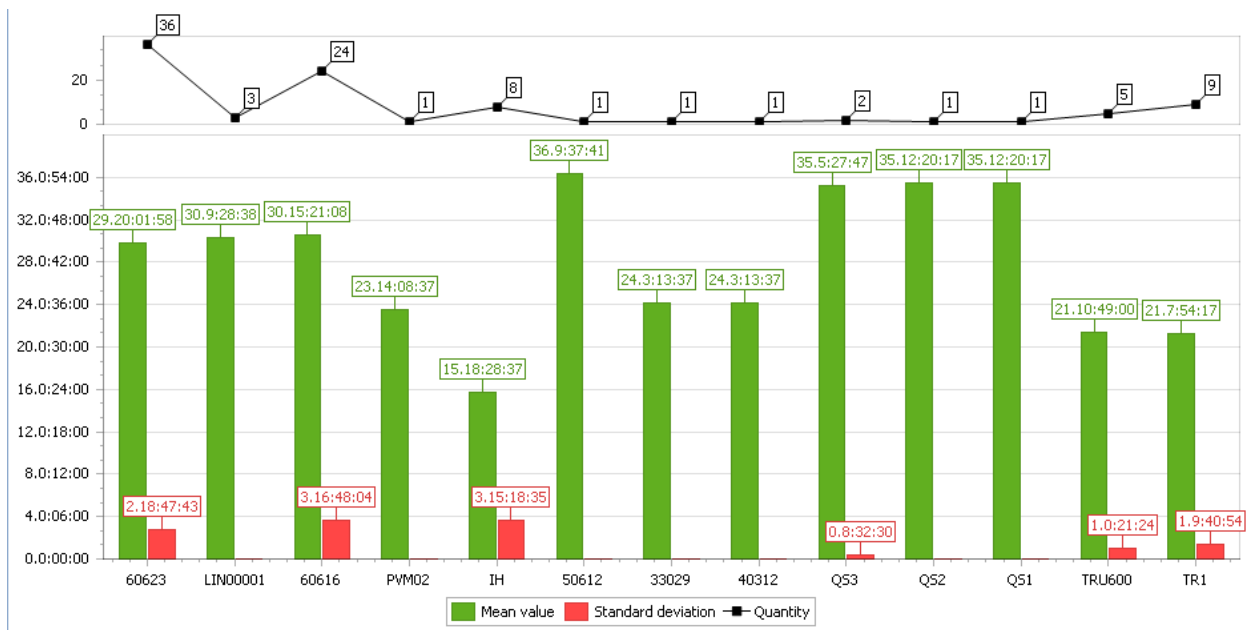
- the mean value AND standard deviation

or

- the total of durations (delays, ...)

are displayed.

Presentation



Data are presented in the format Days:Hours:Minutes:Seconds.

Irrespective of the considered values, the number of included operations is always presented by a line.

Calculations

Mean value

The mean value is calculated as follows:

$$\mu = \frac{\sum_{x=1}^n Earliness_x}{n}$$

$$\mu = \frac{\sum_{x=1}^n Delay_x}{n}$$

$$\mu = \frac{\sum_{x=1}^n Earliness_x + Delay_x}{n}$$

n = Number of OPs produced too early

Standard deviation

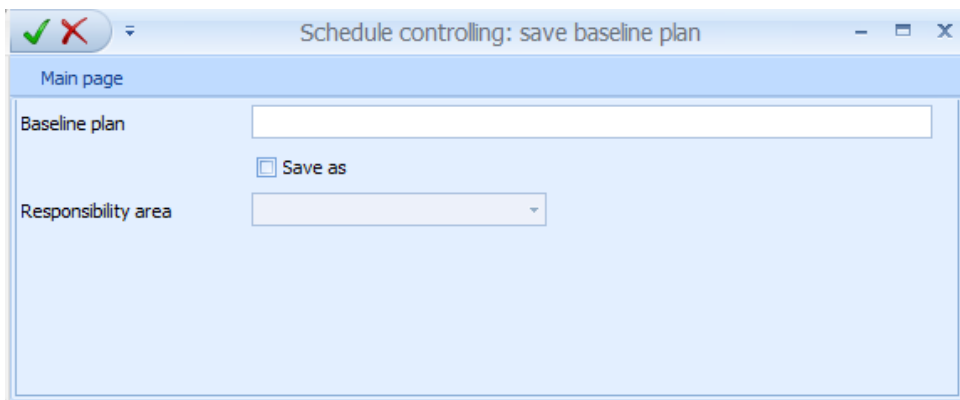
Standard deviation is calculated as follows: $\sigma = \sqrt{Var}$

The variance itself can be calculated for the delay as follows $Var = \frac{\sum_{x=1}^n (Delay_x - \mu)^2}{n}$ (earliness or on-time delivery are calculated in the same way).

Saving of baseline plans

Function authorization: esvb; license: BDE-CAB

The applications “operations” and “pool of orders” provide this function. Baseline plans used for selecting schedule violations can be saved here for selected operations.



The screenshot shows a software window titled "Schedule controlling: save baseline plan". The window has a standard Windows-style title bar with a green checkmark and a red X icon on the left, and minimize, maximize, and close buttons on the right. Below the title bar is a tab labeled "Main page". The main content area contains two input fields: "Baseline plan" and "Responsibility area". The "Baseline plan" field is a text box with a light blue border. Below it is a checkbox labeled "Save as". The "Responsibility area" field is a dropdown menu with a light blue border and a downward arrow on the right.